JAM: Action-Grounded World Pretraining

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Abstract

Video pretraining provides a prior over how scenes evolve. Learning to act also requires a representation of how control relates to that evolution. We study whether action supervision can shape a pretrained world representation through the computation used to generate future observations. JAM, Joint Action–World Modeling, jointly generates future video, motor actions, and visual action consequences. A video backbone and an agent transformer exchange attention keys and values at every layer, allowing action prediction to update the world stream. Visual consequences describe image-space displacement and observability, extending the same training objective to human demonstrations with tracked motion. Independent corruption levels support learning from modalities with different levels of uncertainty. The available training record shows decreasing world, action, and consequence objectives in a completed task adaptation run, while foundation pretraining continues on robot and human data. Controlled comparisons are organized around the effects of information exchange, gradient routing, and consequence supervision on representation quality and transfer. This formulation makes the influence of action learning on world pretraining a concrete, testable property of the model.

1 Introduction

Video models learn regularities in appearance, motion, and scene evolution from visual sequences. These regularities provide useful starting points for embodied learning: a robot must anticipate how objects move, how contact changes a scene, and which futures are consistent with its observations. Recent work transfers video generation to control by extracting actions from imagined trajectories or adapting video models to generate actions directly (Du et al., 2023; Kim et al., 2026). This progress motivates a question about the pretraining process itself: *how should learning to act influence the representation used to predict the world?*

A manipulation video records an interaction through its visible outcome. Robot demonstrations additionally record the commands that accompanied it. These two descriptions constrain each other. The same scene can evolve differently under different commands, and a desired change in the scene can admit several feasible actions. A representation for embodied interaction should preserve these dependencies. Action grounding, in this work, refers to the extent to which a world representation retains information useful for relating commands to their visual consequences. We examine this property through prediction, controlled changes to the inputs, and transfer to new conditions.

We introduce JAM, Joint Action–World Modeling, as a way to expose the world representation to action supervision during pretraining. JAM generates actions and future observations within a shared denoising computation. A pretrained video stream and an agent stream exchange information throughout their depth. Their attention connections also carry gradients, so learning to predict actions can update the features used for world prediction. The central hypothesis is that this reciprocal computation encourages a representation of interaction that remains useful when the environment or task changes. The model makes this hypothesis experimentally accessible through separate controls over information visibility and gradient flow.

The relationship between commands and visible change also offers a route to human video. A motor command is specific to an embodiment and its controller. A visual action consequence describes where tracked points move in the image and whether they remain observable. JAM represents these consequences as generated tokens within the agent stream. Robot and human demonstrations can therefore supervise a common motion interface while retaining their own available annotations. This construction also provides a diagnostic: consequence prediction tests which aspects of interaction are accessible from the learned world representation.

Our study follows the influence of action supervision from the training computation to its downstream use. We first formulate the joint prediction problem and explain how the streams interact. We then describe pretraining with heterogeneous supervision and adaptation for control. The experiments connect optimization evidence to controlled tests of coupling, data composition, and transfer. Together, this organization asks when learning actions and visual consequences changes what a world model represents, and how that change can be measured.

2 Related Work

Video priors for embodied learning. Video generation learns temporal structure from visual data (Wan Team, 2025). UniPi uses generated videos as plans from which actions are recovered (Du et al., 2023). Cosmos Policy incorporates actions, future observations, and values into a pretrained video model through latent frames (Kim et al., 2026). These approaches establish practical routes from generative priors to control. JAM focuses on how action learning influences the video representation during embodied pretraining, with explicit controls over the connections carrying information and gradients.

Joint action and world prediction. Unified World Models (UWM) jointly diffuse actions and observations, sample their noise levels independently, and obtain policies and dynamics models through conditioning (Zhu et al., 2025). JAM builds on this joint-distribution perspective and uses rectified-flow training (Lipman et al., 2023; Liu et al., 2023b). Its study centers on a pretrained video stream coupled throughout its depth to an agent stream. Separating visibility from gradient routing permits comparisons between reading world features and updating those features through action supervision. Conditional sampling provides an additional use of the learned distribution; the representation question motivates the experimental design.

Action representations across embodiments. Robot policy pretraining aligns observations and language with commands across datasets and platforms (Kim et al., 2024; Khazatsky et al., 2024; Walke et al., 2023). Visual trajectories offer another interface to interaction: Any-point Trajectory Modeling learns motion representations for policies, and Track2Act converts predicted point tracks into executable motion (Wen et al., 2024; Bharadhwaj et al., 2024). EgoDex supplies human manipulation video with tracked hand poses (Hoque et al., 2025). JAM incorporates image-space consequences into its joint generative objective, alongside motor commands where available. The corresponding experiments examine command–consequence complementarity and transfer across embodiments under explicitly matched observation and annotation conditions.

3 Joint Action–World Modeling

The design starts from a simple requirement: action prediction and world prediction should be able to use and update each other's representations. A joint objective specifies which variables are generated. The attention structure determines how learning those variables interacts.

3.1 The prediction problem

Let o_0 be the current observation, ℓ a language instruction, and s the available proprioceptive state. We write $h = (o_0, \ell, s)$ for these conditions, with a learned null token for missing state. Let z denote encoded future observations and a a chunk of future motor actions. JAM models $p_\theta(z, a \mid h)$. When visual consequence labels c are available, the same computation models $p_\theta(z, a, c \mid h)$. A fixed video autoencoder maps observations to latents; its decoder maps generated latents back to images. Action and consequence tokens preserve their respective control and image-space meanings.

For each generated modality $m \in \{w, a, c\}$, let $x^w = z$, $x^a = a$, and $x^c = c$. We draw independent standard Gaussian noise ϵ^m and a corruption level τ_m between clean and fully noisy. Rectified-flow training constructs

$$x^m_{\tau_m} = (1 - \tau_m)x^m + \tau_m \epsilon^m, \qquad u^m = \epsilon^m - x^m. \tag{1}$$

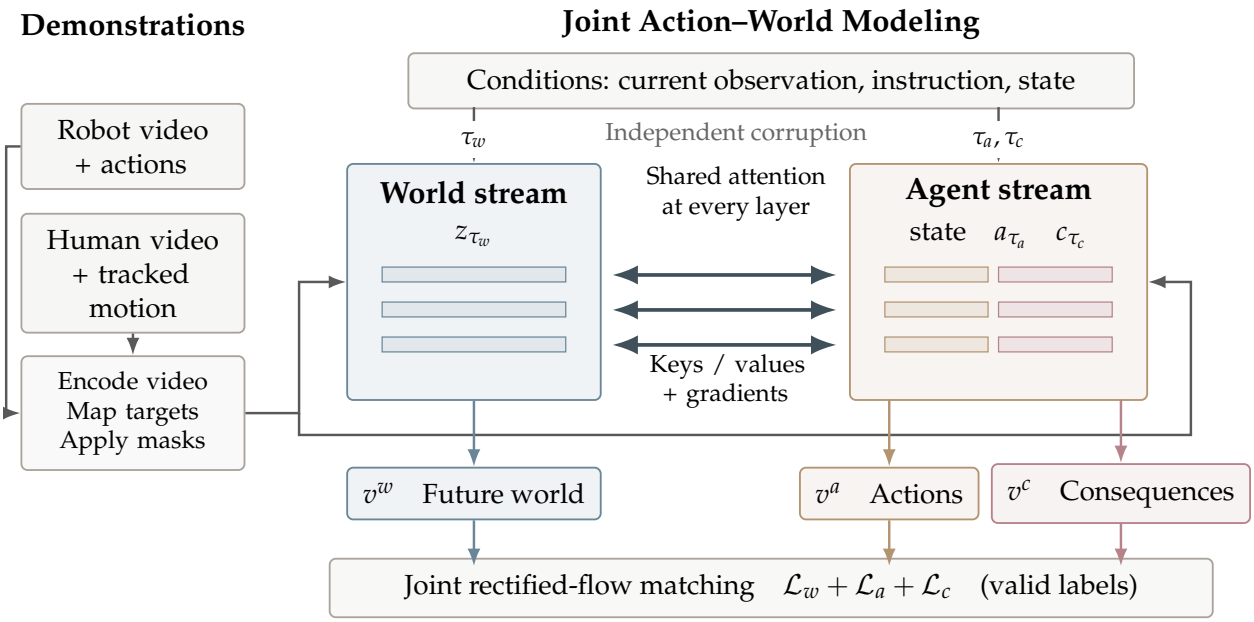


Figure 1: How action learning enters world pretraining. Source adapters encode video and map available commands or tracked motion into masked targets. Independently corrupted future latents, actions, and consequences enter coupled world and agent streams. Attention exchanges carry information and gradients throughout the model. Separate velocity readouts receive their corresponding masked targets within the joint loss. World and agent streams share clean observation and language conditions; the state enters through the agent.

Here u^m is the target velocity along the interpolation from data to noise. The model predicts all available velocities from the corrupted modalities and the clean conditions:

$$(v_\theta^w, v_\theta^a, v_\theta^c) = F_\theta(z_{\tau_w}, a_{\tau_a}, c_{\tau_c}; h, \tau_w, \tau_a, \tau_c). \tag{2}$$

The current configuration samples consequence corruption independently of action corruption. This keeps the confidence of the consequence input distinct from that of the motor input. The formulation follows independent-modality corruption in UWM (Zhu et al., 2025); the coupling described below specifies how the pretrained representations participate.

3.2 Reciprocal computation across streams

The world stream retains a pretrained video transformer's latent embeddings, blocks, and output projection. The agent stream contains a state token followed by action and consequence tokens. It uses separate input projections for each token type and noise-dependent adaptive normalization. Both streams receive the language condition. Their residual widths may differ; their attention projections share a common head geometry.

At each coupled layer, each stream forms its own queries, keys, and values. World queries attend to world and agent keys; agent queries attend to agent and world keys. In schematic form, for stream m and the other stream $\bar{m}$,

$$H'_m = H_m + P_m \ \mathrm{Attn}\big(Q_m, [K_m; K_{\bar{m}}], [V_m; V_{\bar{m}}]\big), \tag{3}$$

where H_m is the layer input, P_m maps attention outputs to its residual width, and brackets denote token concatenation. Each block then applies language cross-attention and a feed-forward update. Consequence tokens belong to the agent computation and participate in the same exchange. Figure 1 shows the complete training flow.

The default coupling is bidirectional at every layer, with the exchanged tensors attached to the computation graph. Thus action loss has a route into world parameters through the world keys and values read by the agent; world loss has a reciprocal route into agent parameters. These routes establish a mechanism for

action supervision to influence the world representation. Their value for transfer is an empirical question. Visibility masks and gradient stops expose the mechanism to controlled comparison.

The current observation forms a clean prefix in the world sequence. Prefix queries attend only to the clean prefix, while generated world tokens can read the prefix and the agent stream. This preserves an observation context throughout joint prediction. The state token is a clean input to the agent stream and is excluded from the generation loss.

3.3 Learning under partial supervision

Each example carries coordinate masks M^m for the labels it supplies. We use a masked mean squared error, MSE_{M^m}, averaged over valid coordinates within that example. Let $\mathcal{B}_m$ be the batch examples with at least one supervised coordinate for modality m. The implemented objective is

$$\mathcal{L} = \sum_{m \in \{w,a,c\}} \lambda_m \frac{1}{|\mathcal{B}_m|} \sum_{i \in \mathcal{B}_m} \omega(\tau_{m,i})\ \mathrm{MSE}_{M_i^m}\left(v_{\theta,i}^m, u_i^m\right), \tag{4}$$

where λ_m controls each modality's contribution and ω weights its noise level. An empty supervised set contributes zero. World loss covers future latents; the clean prefix is excluded. Missing coordinates are masked after corruption as well as in the loss. Robot batches supervise actions and video, while human batches supervise video and the available consequences. Consequence annotations may also accompany robot data. Appendix A gives the implemented sampling distribution and normalization rules.

4 Action-Grounded World Pretraining

Joint prediction provides the learning mechanism. Its use across demonstrations depends on preserving what each source actually observes and controls. We therefore align the meaning of the inputs before combining their supervision.

4.1 Commands and their visual consequences

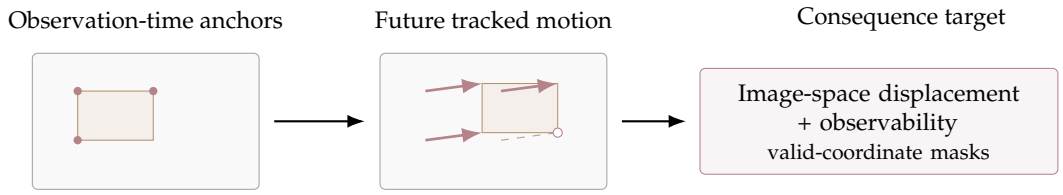
A source adapter maps motor commands into a typed layout for end-effector pose, gripper aperture, joints, and additional embodiment-specific coordinates. The adapter also records which coordinates exist. Source-specific normalization is applied after this mapping. Rotation coordinates retain their geometric convention, and gripper aperture follows a shared closed-to-open convention. This interface preserves physical meaning when sources populate different subsets of the layout.

Visual action consequences describe displacement from observation-time anchors. For keypoint k at future time t, let $q_{t,k}$ be its image-normalized position, $q_{0,k}$ its anchor, and $b_{t,k}$ its observability. The target is

$$c_{t,k} = \left[g(q_{t,k} - q_{0,k}),\ 2b_{t,k} - 1\right], \tag{5}$$

where g is a fixed displacement gain. Tracked actuator points use their observation-time position as the anchor; scene probes use their query location in that frame. Masks distinguish missing slots from observed low visibility. The resulting sequence is projected into consequence tokens and trained by the same velocity objective as actions and video.

Figure 2 illustrates this interface. Point trackers provide one route to such labels (Karaev et al., 2024). The active human-data adapter instead projects supplied hand poses through the recorded camera calibration and combines their confidence with visibility (Hoque et al., 2025). These demonstrations supply tracked motion while motor commands remain unobserved. Consequences express visible motion in a common coordinate format; camera motion, projection, and embodiment still affect their distribution. Cross-embodiment invariance is therefore evaluated through held-out transfer.



Schematic: filled points are visible; the open point illustrates an occluded track.

Figure 2: What consequence supervision represents. Motion is expressed relative to an anchor in the current image. Visibility is predicted separately from displacement validity, so an occluded point supplies a visibility target while its displacement is masked. The active human adapter obtains tracks by projecting supplied hand poses; robot annotations can combine actuator tracks and scene queries. This is an explanatory drawing, not a model prediction.

4.2 Pretraining from heterogeneous demonstrations

Each training window contains an instruction, a current observation, future frames, and the available state, action, and consequence annotations. Camera views occupy fixed positions in an image canvas, which is encoded by the frozen video autoencoder. A frozen text encoder supplies language embeddings. Learned input projections route control and consequence coordinates into the agent stream. The pretrained world transformer and the new agent transformer are optimized together.

We sample sources using a temperature applied to their numbers of valid training windows. Each micro-batch contains a single source so its camera geometry and annotation pattern remain consistent. Robot and human batches enter the same optimization stage. Human data thus contributes a world objective and a consequence objective while robot data anchors the command interface. The current recipe and training budget appear in Section 5; larger acquired corpora enter the accounting when their caches are incorporated into a run.

Increasing data, updates, and model capacity answers different scaling questions. The study varies each separately: subsets of a fixed corpus test data diversity, checkpoint milestones test training duration, and agent capacity tests the cost of the learned control interface. The key comparison preserves robot exposure when adding human data, followed by a compute-matched comparison. This separates an additional supervision source from an additional optimization budget.

4.3 Adaptation and conditional use

Downstream adaptation retains the joint objective and replaces the data mixture with demonstrations for the target environment. At execution time, JAM starts the generated streams from noise, integrates predicted velocities toward clean samples, and executes a prefix of the action chunk before observing again. The world decoder and consequence readout make the jointly sampled future available for diagnostics.

The joint formulation also permits conditioning on observed generated variables. Given an action chunk, world prediction estimates its associated visual evolution; given a future observation, inverse prediction estimates compatible actions. Independent noise levels expose the model to varying uncertainty in these inputs. Practical conditional samplers additionally specify which variables are supplied, which remain latent, and how each latent variable is integrated. The primary implemented policy uses a shared decreasing sampling schedule, while conditional-sampling comparisons remain part of the evaluation protocol. The current objective remains focused on world, action, and consequence prediction.

5 Experiments

The evaluation asks whether reciprocal action–world learning changes representations in ways that support control and transfer. Its ordering follows the proposed mechanism: establish optimization, isolate the coupling, vary the supervision, and examine the resulting behavior under distribution shift.

Active source	Training windows	Sampling	Supervision
DROID	1,674,648	50.9%	World + action
BridgeData V2	454,690	20.4%	World + action
EgoDex	738,550	28.7%	World + consequence

Table 1: Which data contributes which signal? Active cached windows after the episode holdout, as recorded in the project ledger. Probabilities apply temperature 0.7 to window counts; window duration follows each source's native frame rate. The recipe is shared by the archived foundation-training segment. Additional acquired sources remain outside this table.

5.1 Experimental setup

Evidence status. This manuscript records the project snapshot of 7 September 2026. Task adaptation on LIBERO completed 20,000 updates. Foundation pretraining is active; the archived log reaches 5,200 updates. The available artifacts establish training behavior. Closed-loop evaluation and controlled representation comparisons are awaiting complete measurements. Tables distinguish *measured training statistics*, *externally reported results*, and *layout targets*, with the latter marked by superscript T throughout. Target values support manuscript layout only. Artifact hashes and the claim ledger accompany the source.

Data and splits. The active pretraining mixture contains DROID (Khazatsky et al., 2024), BridgeData V2 (Walke et al., 2023), and EgoDex (Hoque et al., 2025); Table 1 records the cached training windows and sampling probabilities. The cache reserves an episode-level holdout using a deterministic one-in-ten partition. All windows from an episode follow that partition. The active mixture contains the training portions of these sources and excludes the benchmark datasets below. Holdout windows serve open-loop checks; environment resets define the separate closed-loop test sets. Dataset expansion is tracked separately from data already consumed by training.

Model and optimization. The world stream uses Wan2.2-TI2V-5B through its released Diffusers implementation, initialized with public video weights. The agent stream is trained from scratch. The combined model has approximately 6.02 billion parameters, including a 1.02-billion-parameter agent with width 1,024 and 30 layers. Both streams use 24 attention heads of dimension 128. The video autoencoder and text encoder remain frozen. Foundation training uses eight H200 accelerators, a batch of 16 per device, and four accumulation steps, giving a global batch of 512. AdamW uses world/agent learning rates of 0.00005/0.0001, momentum coefficients 0.9/0.95, weight decay 0.01, and gradient clipping at 1. The planned budget is 60,000 updates with 1,500 warmup updates and a final 6,000-update decay. All three loss weights are 1. The resolved configuration uses independent corruption levels and zero explicit boundary-sampling probability. The completed direct-adaptation run uses four H200 accelerators, global batch 64, and a cosine learning-rate schedule; Appendix A records the remaining settings.

Tasks and comparison protocol. The core downstream arenas are LIBERO (Liu et al., 2023a), RoboTwin 2.0 Full (Chen et al., 2025), VLABench (Zhang et al., 2025), and the RoboCasa365 target-task split (Nasiriany et al., 2026). Direct adaptation starts from the public video initialization plus a fresh agent; pretrained adaptation starts from a JAM foundation release. The paired comparison fixes the demonstrations, action adapter, training budget, sampler, and environment version. LIBERO-Plus (Fei et al., 2025) and LIBERO-PRO (Zhou et al., 2025) supply robustness tests. RoboTwin clean-to-randomized transfer is evaluated separately from training on its Full mixture. The proposed final protocol uses three training seeds, 50 trials per LIBERO task, and the official one-trial-per-variant LIBERO-Plus enumeration. Each remaining suite follows its pinned task and reset manifest. Full-suite external averages remain distinct from the current task-subset experiments.

Metrics and statistics. Control is measured by task success rate, with subtask progress retained where supplied by the benchmark. World prediction uses perceptual image distance and tracked-motion error. Consequence prediction uses average displacement error in image-normalized coordinates and visibility calibration. Frozen-world probes predict actions and consequences from intermediate world features with generated action and consequence inputs fully corrupted. A shuffled-label probe and an observation-only predictor establish reference levels. Planned uncertainty estimates use a paired hierarchical bootstrap over training seeds, tasks, and shared episode resets with 10,000 resamples and 95% intervals. Open-loop metrics use episode-level resampling.

Objective	First logged batch	Final-window mean
World	0.4221	0.0573
Action	1.3282	0.0108
Consequence	1.7038	0.0305
Total	3.4541	0.0985

Table 2: Measured joint optimization. LIBERO direct adaptation, 20,000 updates. The final column averages the last 10% of logged points using the archived training summary. These are training losses. All values come from `e0_libero_bi_summary.json`; the initialization contains public video weights and a fresh agent.

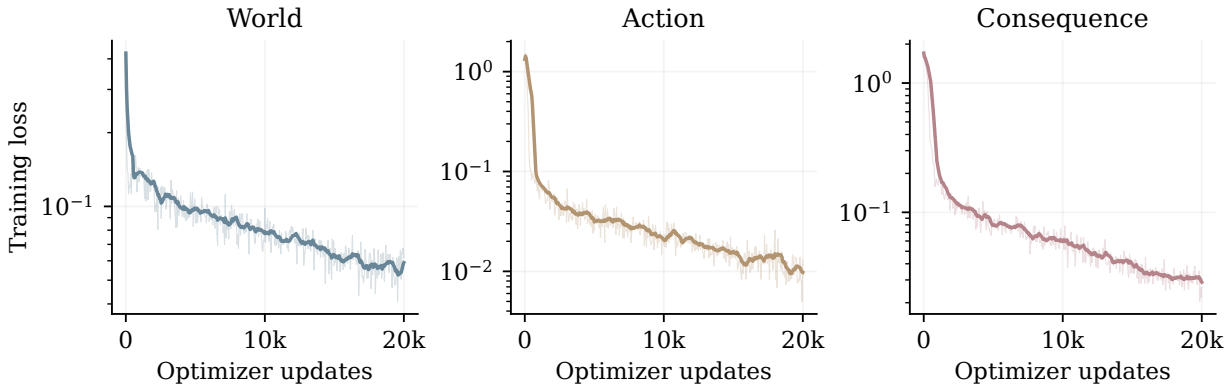


Figure 3: Do the implemented objectives optimize together? Measured training losses from the completed LIBERO direct-adaptation run. Thin lines are logged values; thick lines are trailing means over 11 logged points. Each panel retains its own objective scale. The figure reports optimization on training batches; held-out control and representation evaluations provide the separate tests of transfer. Source: `artifacts/measurements/e0_libero_bi.jsonl`.

Initialization	LIBERO	RoboTwin	VLABench	RoboCasa365
External results, verbatim from R1; source protocols				
R1	99.3	93.60	58.9	38.2
Planned paired comparison; JAM values are layout targets				
Direct adaptation	97.2^{T}	74.5^{T}	24.8^{T}	12.4^{T}
JAM pretraining + adaptation	99.4^{T}	94.6^{T}	59.8^{T}	39.6^{T}

Table 3: Does embodied pretraining improve downstream control? Success percentages. The R1 row reproduces the source averages of external reference R1 exactly; these are measurements of that external system. JAM rows are planned layout targets. The external RoboCasa365 value is a full source average, whereas the current JAM protocol targets 50 tasks; protocol alignment is required for a direct comparison. The reference row therefore supplies context rather than a matched ranking. Superscript T marks layout targets awaiting measurement. These values specify the display layout and carry no empirical claim.

5.2 Optimization and downstream control

Table 2 and Figure 3 report the completed joint-optimization record, while Table 3 separates external control results from the planned paired downstream comparison.

5.3 Which part of joint learning matters?

Table 4 defines the comparison that separates world-only and action-only learning, independent multitask prediction, auxiliary supervision, and reciprocal joint computation.

Pretraining computation	LIBERO	LIBERO-Plus	Action probe
World-only pretraining	94.0^{T}	42.0^{T}	0.3^{T}
Action-only pretraining	95.0^{T}	45.0^{T}	0.32^{T}
Independent multitask	96.1^{T}	47.6^{T}	0.38^{T}
Auxiliary, detached world features	96.8^{T}	49.4^{T}	0.4^{T}
World-to-agent, attached	97.0^{T}	51.7^{T}	0.46^{T}
JAM, reciprocal and attached	97.2^{T}	53.8^{T}	0.52^{T}

Table 4: What does reciprocal learning add? Planned comparison after the same downstream adaptation. The first two columns are success percentages; the probe reports R-squared. All arms receive matched clean conditions and data. Auxiliary supervision reads detached world features; the attached one-way arm permits action gradients into the world stream. Reciprocal coupling adds agent information to world queries. Appendix B specifies the condition-matching requirement. Superscript T marks layout targets awaiting measurement. These values specify the display layout and carry no empirical claim.

Pretraining data	LIBERO	LIBERO-Plus	Consequence ADE
Robot data	98.4^{T}	74.2^{T}	0.12^{T}
Robot + human video	98.9^{T}	82.6^{T}	0.1^{T}
Robot + human video and consequences	99.4^{T}	90.2^{T}	0.08^{T}

Table 5: What does human motion supervision contribute? Planned data-composition comparison with matched robot exposure. The human-video arm masks the consequence objective. The final arm includes projected hand-motion targets. Success is in percent; average displacement error (ADE) uses image-normalized coordinates on the same held-out episode list. A companion compute-matched run and a label-density sweep separate supervision from additional updates. Superscript T marks layout targets awaiting measurement. These values specify the display layout and carry no empirical claim.

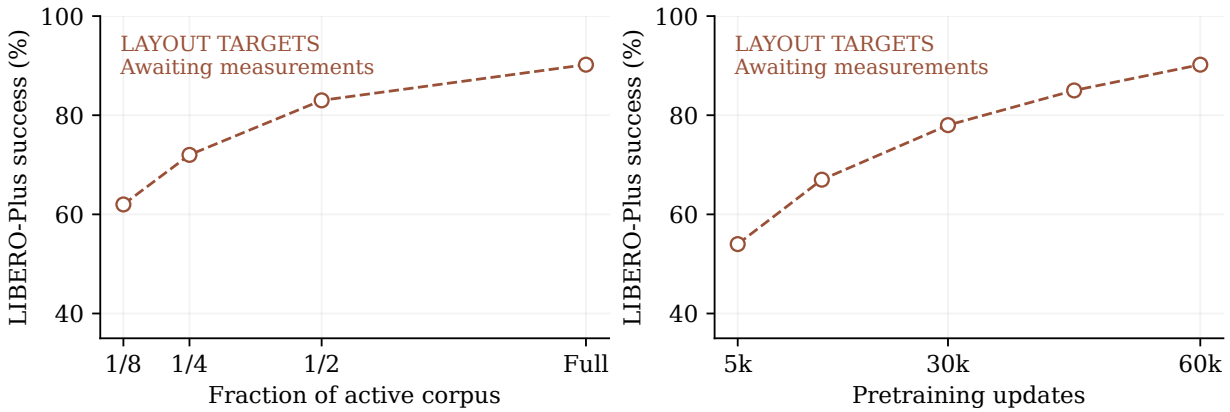


Figure 4: How should a pretraining gain vary with data and training duration? Layout targets for LIBERO-Plus success under a fixed downstream protocol. The left panel varies a stratified subset of the active corpus at fixed updates; the right panel varies updates at fixed corpus. Every plotted point is a PLANNED-VALUE awaiting measurement. The curves illustrate the display only; their slopes and ordering carry no empirical claim.

5.4 Data composition and scaling

Tables 5 and 8, together with Figure 4, specify how robot exposure, human video, consequence supervision, and optimization duration enter the scaling study.

5.5 Out-of-distribution generalization and transfer

Table 6 organizes the planned evaluations by the shift applied to a fixed adapted checkpoint.

Shift and evaluation	Direct	Pretrained
Visual and layout shift: LIBERO-Plus	53.8^{T}	90.2^{T}
Initial-state shift: LIBERO-PRO	26.0^{T}	46.8^{T}
Task-instruction shift: LIBERO-PRO	6.0^{T}	12.6^{T}
Environment shift: RoboTwin randomized	14.2^{T}	62.4^{T}
Category shift: VLABench	16.0^{T}	39.0^{T}
Embodiment transfer: held-out platform	28.0^{T}	42.0^{T}

Table 6: Where does pretraining transfer? Planned success percentages under paired evaluation resets. RoboTwin reports the randomized test partition after clean-only adaptation, distinct from Full-mixture training. Held-out embodiment transfer requires an explicitly excluded platform, a frozen pretraining split, and the same downstream demonstration budget for both initializations. Each row is a separate protocol rather than a shared aggregate. Superscript T marks layout targets awaiting measurement. These values specify the display layout and carry no empirical claim.

Diagnostic	Comparison A	Comparison B
World action probe, R-squared	0.4^{T}	0.52^{T}
World consequence probe, ADE	0.12^{T}	0.08^{T}
Paired action intervention, ADE	0.18^{T}	0.1^{T}
World prediction, perceptual distance	0.24^{T}	0.2^{T}
Predicted-consequence condition, ADE	0.1^{T}	0.13^{T}

Table 7: Which interaction information is represented? Layout targets. Rows 1–3 compare detached auxiliary supervision (A) with reciprocal coupling (B). Row 4 compares action conditioning (A) with action plus observed consequence conditioning (B). Row 5 compares observed (A) with predicted (B) consequences, retaining a possible degradation in the planned display. ADE denotes average displacement error. Probes fully corrupt generated inputs; interventions pair alternative commands with simulator rollouts from the same initial state. Superscript T marks layout targets awaiting measurement. These values specify the display layout and carry no empirical claim.

5.6 Mechanism and representation

Table 7 specifies the probes, matched interventions, and consequence-conditioning comparisons used to examine what the world representation retains.

6 Conclusion

JAM places action prediction inside the computation used to learn world dynamics. Its coupled streams give action supervision a direct gradient route into a pretrained video representation, while visual consequences provide a motion target shared by robot and human demonstrations. The completed adaptation record shows that the implementation can optimize world, action, and consequence prediction together. Foundation training and the controlled evaluation address the remaining scientific question: when does this coupling produce a representation that transfers to new interactions? The present evidence supports the training formulation and motivates measuring representation change separately from optimization progress.

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Appendix

A Implementation and training details

Prediction horizons and coordinates. The current model predicts 32 action steps in an 80-dimensional typed layout. Consequences use 16 steps with 48 available keypoint slots, each containing two displacement coordinates and one observability coordinate. The agent sequence therefore contains 49 tokens: one state token, 32 action tokens, and 16 consequence tokens. Slots are shared storage positions with source-specific validity masks. The current LIBERO adapter fills 39 consequence slots; EgoDex fills 14 hand-point slots. Unused slots carry a false mask.

Actions use source adapters that preserve the physical meaning of pose, aperture, and joint coordinates. Rotation is represented by the first two columns of its rotation matrix, concatenated in column order, and bypasses statistical scaling. Gripper aperture is first expressed from closed to open in the interval $[0, 1]$ and then mapped to $[-1, 1]$. The active robot recipe uses per-source first and ninety-ninth percentiles for the remaining valid coordinates, with clipping to $[-1, 1]$. Min–max and standard-score normalization are supported alternatives; the archived runs identify the active choice. Displacement uses image-normalized coordinates with gain 4. Observability remains a separate target, including for occluded points whose displacement is masked.

Temporal and visual inputs. A window selects nine frames at a stride of four source steps, spanning 33 time indices. The video autoencoder produces a clean observation latent followed by future latents. The image canvas is 384×256 pixels for the two-view template and 384×320 for the three-view template. The template reserves positions for unavailable views rather than inventing additional observations. DROID, Bridge, and EgoDex retain native frame rates of 15, 5, and 30 Hz in the resolved recipe. Consequently, a step horizon corresponds to different elapsed durations across sources. Duration-matched comparisons are a separate control for cross-source transfer.

Noise sampling and weighting. For a uniform random variable u and shift r, the implemented level is $\tau = ru/(1 + (r - 1)u)$. The world and action shifts are 5 in the archived runs; the independent consequence draw uses the action shift. The bell-shaped weight is proportional to $\exp[-2(\tau - 1/2)^2] - \exp(-1/2)$, normalized by its mean under the shifted sampling distribution. World, action, and consequence levels are independent during training. The default sampling path decreases all three levels together with 10 Euler steps. Current training uses no additional probability mass on clamped boundary conditions. A boundary-mixture comparison is planned independently of the primary coupling comparison.

Completed adaptation and active pretraining. The completed LIBERO run uses 20,000 optimizer updates, batch 16 per device on four H200 accelerators, one accumulation step, and seed 0. World and agent learning rates both start at 0.0001 with 1,000 warmup updates and cosine decay to 0.01 of the peak. The foundation run uses the settings in Section 5, with a warmup–stable–decay schedule ending at 0.05 of the peak. Training uses bfloat16 computation, float32 gradient reduction, and activation checkpointing. Foundation training uses hybrid sharding across two nodes of four accelerators. The video backbone is initialized from released video weights; embodied foundation training is performed within JAM.

The archived foundation configuration contains DROID, Bridge, and full EgoDex from its first optimizer update. An earlier robot-only stage pointer was replaced before training began. Checkpoints are saved every 300 updates. The first retained milestone release is at update 5,100, and the archived log extends to 5,200. The intended downstream launch point is the 15,000-update foundation milestone. These milestones describe training duration rather than completed scaling comparisons. The completed LIBERO adaptation has 401 logged points; its final-window summary averages the last 10% of those points. A zero consequence loss in a robot-only microbatch indicates absent supervision, so foundation summaries condition on the recorded supervised fraction.

B Controls required by the scientific question

Separating information and gradients. World-only pretraining supervises future video. Action-only pretraining supervises commands. Independent multitask pretraining supervises both with separate

Planned capacity or annotation setting	LIBERO-Plus success
Agent width 768; full action labels	85.0^{T}
Agent width 1024; full action labels	90.2^{T}
Agent width 1280; full action labels	91.0^{T}
Action-label density 25 percent	82.0^{T}
Action-label density 50 percent	86.0^{T}
Action-label density 100 percent	90.2^{T}

Table 8: Which scale axis matters? Layout targets in percent. Agent-width comparisons keep the video backbone, head geometry, depth, data, and update budget fixed and report the resulting compute separately. Annotation-density comparisons keep all video windows and robot exposure fixed while masking action labels at the episode level. The full-label setting is the shared reference for the density sweep. These settings require dedicated source configurations before execution. Superscript T marks layout targets awaiting measurement. These values specify the display layout and carry no empirical claim.

generated-stream computations. Auxiliary supervision reads detached world features. The attached one-way arm uses the same visibility while allowing action gradients through world keys and values. Reciprocal JAM additionally exposes agent keys and values to world queries. Each pretraining arm receives the same source sequence and then the same downstream adaptation procedure. Direct adaptation from public video weights supplies the separate baseline for the value of embodied pretraining.

Strict isolation of coupling requires identical clean conditions in every arm. In the audited implementation, disabling agent visibility also removes the state token from the world stream's attention. The existing visibility switches therefore change both generated-token exchange and part of the conditioning interface. The planned comparison routes equivalent static state and current-observation information to every arm before interpreting a performance difference as an effect of coupling. The paper's target table describes this stricter protocol, whose results remain awaiting measurement. It does not treat the existing switches as an already completed controlled experiment.

Representation probes and interventions. A world probe freezes the pretrained parameters and reads the same selected layer and pooling rule in every arm. Generated action and consequence tokens are fully corrupted during feature extraction; their clean targets are used only to train or evaluate the probe on separate episodes. A linear probe tests accessible information, while a capacity-matched nonlinear probe checks dependence on readout capacity. Observation-only and shuffled-label predictors provide reference levels. World features are compared at fixed noise levels and matched visible context.

Action shuffling tests sensitivity to the action input. Counterfactual prediction additionally requires a target future obtained by executing the alternative action from the same simulator state. The intervention protocol therefore saves a reset state, executes each admissible command sequence, and scores each generated future against its matching rollout. Corruption tests vary timing and command magnitude within valid control limits. This distinguishes sensitivity from correct prediction of an intervention.

Consequence conditioning compares commands alone, observed consequences alone, and both together on the same test windows. Predicted consequences form a separate condition that retains their model error. The resulting gap measures the cost of prediction and any distribution shift at the conditioning interface. A held-out-embodiment study excludes the target platform from pretraining and fixes its downstream demonstration budget. Camera configuration, temporal duration, and available keypoint groups accompany each result.

Conditional sampling checks. Every conditional use specifies all generated variables. For action-conditioned world generation, an unobserved consequence stream remains latent and must be denoised on its own schedule. The audited action-clamped sampler couples its default consequence schedule to the action schedule; supplying only actions can therefore leave an initialized consequence latent unchanged. Conditional benchmark measurements require this behavior to be resolved or all clamped variables to be supplied explicitly. The manuscript consequently treats policy sampling as the primary implementation and conditional comparisons as pending validation. Independent training levels define the corruption family; each deployment sampler still requires its own correctness and fidelity check.

External evaluation	Reported success (percent)
Task control: LIBERO	99.3
Bimanual control: RoboTwin Full	93.60
Perturbations: LIBERO-Plus	69.2
Clean-to-randomized: randomized only	48.7
Clean-to-randomized: source mean	69.0
Language and task transfer: VLABench	58.9
Household tasks: RoboCasa365	38.2
Manipulation coverage: RoboDojo	11.92
Mobile manipulation: EBench	49.4
Humanoid setup: RoboCasa-GR1	60.5

Table 9: External reference anchors. All cells are reproduced verbatim from external reference R1, with the source's own training and evaluation protocols. In particular, 69.0 is the clean-to-randomized average of clean and randomized performance; its randomized-only score is 48.7. The machine-readable source pointer for every value is retained in `artifacts/external_reference.json`. These values describe the external system only.

C Evidence boundaries and failure analysis

The evidence package contains the completed adaptation log, its result summary, a fixed prefix of the ongoing foundation log, and sanitized resolved configurations. The public manifest records source paths, file hashes, the audit revision, and the status of each artifact. Local asset paths are replaced by logical references in the published configurations; original-file hashes preserve the connection to the inspected source. These configuration snapshots document the runs and are not standalone launch configurations.

The LIBERO closed-loop evaluation was interrupted before a complete suite result file was produced. Partial rollout videos therefore contribute neither success rates nor selected qualitative evidence to the paper. Foundation transfer, coupling ablations, and consequence probes await completed runs. External reference results retain their own provenance and evaluation scope. The planned tables and scaling figure use design values inherited or extended from the working manuscript; they are layout targets rather than preregistered outcomes or model predictions.

The data audit identified extreme displacement labels in a subset of human-video windows. A subsequently added displacement mask is configured to reject values beyond magnitude 8; the first archived training segment precedes activation of this correction. Its losses retain the original label distribution. Evaluating the filter requires a separately identified segment and comparison. This example motivates tracking label confidence and projection failure alongside prediction error. Camera motion, occlusion, and the mismatch between observed and predicted consequence distributions are additional failure settings in the evaluation design.

D External reference results

Table 9 reorganizes source-reported results around the capabilities they evaluate. The linked source identifies the external method and retains its original table entries. R1 is a provenance identifier used to distinguish that system from JAM; the artifact includes an immutable source URL and an exact source pointer for every copied row.

E Reproduction and experiment ledger

Measured logs, external references, and layout targets occupy separate artifact files. The README gives the complete build procedure. The commands `python3 scripts/build_artifacts.py` and `python3 scripts/validate.py` regenerate the numerical displays and check evidence consistency, respectively.

The protocol ledger links each scientific question to inspected source configurations and required result artifacts. It identifies controls that still require implementation changes. Promoting a target to a measurement requires its result file, checkpoint, evaluation manifest, and aggregation command; the corresponding scientific claim is then reviewed against that evidence.